

PHYS414/514 Project: Stars, from Newton to Einstein

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I. NEWTON

a) We define $A\xi = r$ since ξ is the scaled radius. We assume polytropic equation of state and take pressure as $P = K\rho^{1+\frac{1}{n}}$. We insert this into our ODE

$$\frac{dp(r)}{dr} = \frac{-Gm(r)\rho(r)}{r^2} \quad (1)$$

which we obtained from the change in pressure between radii and get

$$K \frac{d(\rho^{1+\frac{1}{n}})}{dr} = K(1 + \frac{1}{n})\rho(r)^{\frac{1}{n}} \frac{d\rho(r)}{dr} = \frac{-Gm(r)\rho(r)}{r^2} \quad (2)$$

Rearranging the equation, we get

$$\frac{-K\rho(r)^{\frac{1}{n}-1}(1 + \frac{1}{n})r^2 \frac{d\rho(r)}{dr}}{G} = \frac{-Kr^2(n+1) \frac{d(\rho(r)^{\frac{1}{n}})}{dr}}{G} = m(r) \quad (3)$$

Since we know

$$\frac{dm(r)}{dr} = 4\pi r^2 \rho(r) \quad (4)$$

we take the derivative of the rearranged equation with respect to r

$$\frac{dm(r)}{dr} = \frac{-K(n+1)}{G} \frac{d}{dr} \left(\frac{r^2 d(\rho(r)^{\frac{1}{n}})}{dr} \right) = 4\pi r^2 \rho(r) \quad (5)$$

$$\frac{K(n+1)}{4\pi G} \frac{1}{r^2} \frac{d}{dr} \left(\frac{r^2 d(\rho(r)^{\frac{1}{n}})}{dr} \right) + \rho(r) = 0 \quad (6)$$

Taking $a^2 = \frac{K(n+1)\rho_c^{\frac{1}{n}-1}}{4\pi G}$ and substituting $\frac{K(n+1)}{4\pi G}$ to $a^2 \rho_c^{1-\frac{1}{n}}$, we get

$$a^2 \rho_c^{1-\frac{1}{n}} \frac{1}{r^2} \frac{d}{dr} \left(\frac{r^2 d(\rho(r)^{\frac{1}{n}})}{dr} \right) + \rho(r) = 0 \quad (7)$$

Substituting $\frac{a^2}{r^2}$ to $\frac{1}{\xi^2}$, $\rho(r)$ to $\rho_c \theta^n$, $\frac{d}{dr}$ to $\frac{1}{a} \frac{d}{d\xi}$, and dividing the equation by ρ_c , we get

$$\frac{1}{\xi^2} \frac{d(\xi^2 \frac{d\theta}{d\xi})}{d\xi} + \theta^n = 0 \quad (8)$$

which is Lane-Emden equation

Solving the equation around $\xi = 0$, we get

$$\theta_1(\xi) = 1 - \frac{1}{6}\xi^2 + \frac{n}{120}\xi^4 + \dots$$

and

$$\theta_2(\xi) = \frac{1}{\xi} - \frac{\xi}{2} + \frac{\xi^3}{24} + \dots$$

We do not consider θ_2 since it is not physical. We can see that $\theta'(0) = 0$.

For $n=1$, $\theta(\xi) = \frac{\sin \xi}{\xi}$

To find the total mass M, we write

$$M = \int_0^R 4\pi r^2 \rho(r) dr \quad (9)$$

which is equal to

$$\int_0^{\xi_1} 4\pi a^3 \rho_c \xi^2 \theta^n(\xi) d\xi \quad (10)$$

Substituting θ^n ,

$$-4\pi a^3 \rho_c \int_0^{\xi_1} \frac{d(\xi^2 \frac{d\theta}{d\xi})}{d\xi} d\xi \quad (11)$$

We conclude that

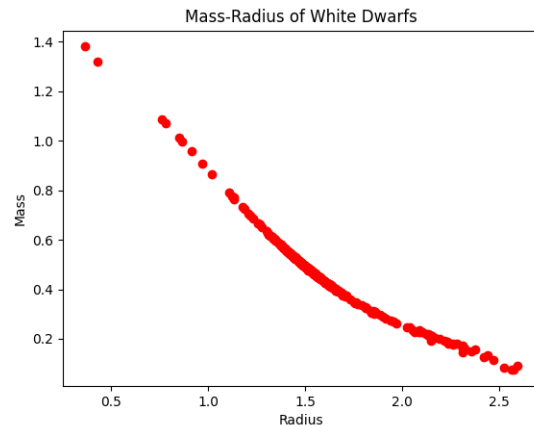
$$M = 4\pi R^3 \rho_c \left(-\frac{\theta'(\xi_1)}{\xi_1} \right) \quad (12)$$

By getting ρ_c from equation $a^2 = \frac{K(n+1)\rho_c^{\frac{1}{n}-1}}{4\pi G}$ and inserting into equation 12, we get

$$M = (4\pi)^{\frac{1}{1-n}} R^{\frac{3-n}{1-n}} \left(\frac{-\theta'(\xi_1)}{d\xi} \right) \left(\frac{K}{G} \right)^{\frac{n}{n-1}} (n+1)^{\frac{n}{n-1}} \xi_1^{\frac{1+n}{n-1}} \quad (13)$$

so $M \propto R^{\frac{3-n}{1-n}}$.

b)



c)

$$P = C[x(2x^2 - 3)(x^2 + 1)^{\frac{1}{2}} + 3 \sinh^{-1}(x)], x = \left(\frac{\rho}{D}\right)^{\frac{1}{q}}$$

When $x \ll 1$, using Taylor expansion, we get

$$(x^2 + 1)^{1/2} = 1 + \frac{x^2}{2} - \frac{x^4}{8} + \dots \quad (14)$$

$$\sinh^{-1}x = x - \frac{x^3}{6} + \frac{3x^5}{40} + \dots \quad (15)$$

$$P = C\left[\frac{8x^5}{5} + \dots\right] \quad (16)$$

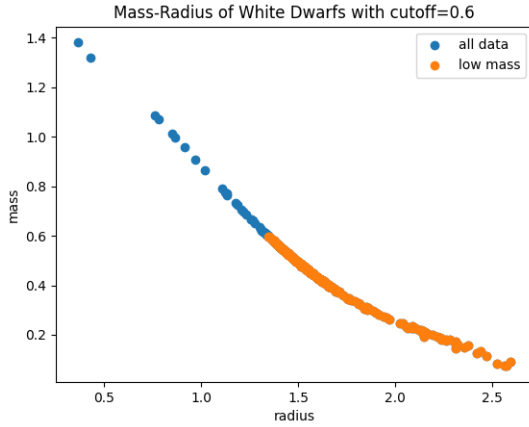
We end up with $P = K_* \rho^{1 + \frac{1}{n_*}}$
 where $n_* = \frac{q}{5-q}$ and $K_* = \frac{8C}{5D^{\frac{5}{q}}}$

e)

$$\nu' = \frac{2M}{r(r - 2M)} \quad (17)$$

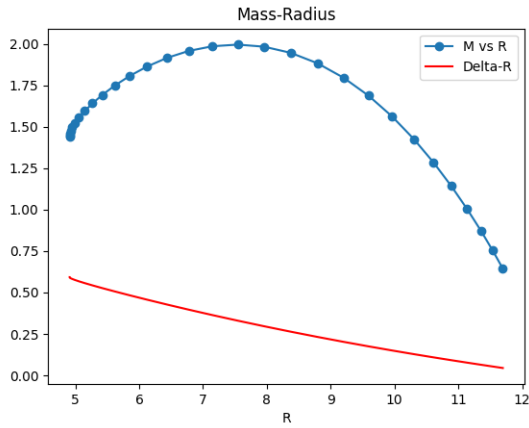
$$\int_R^r \nu' dr' = \int_R^r \frac{2M}{r'(r' - 2M)} dr' \quad (18)$$

$$\nu(r) - \nu(R) = \int_R^r \frac{2M}{r'^2} \frac{1}{1 - \frac{2M}{r'}} dr' \quad (19)$$



II. EINSTEIN

a,b)



$$\nu(r) - \nu(R) = \int_{1 - \frac{2M}{r}}^{1 - \frac{2M}{R}} \frac{1}{u} du \quad (20)$$

$$\nu(r) = \ln\left(1 - \frac{2M}{r}\right) - \ln\left(1 - \frac{2M}{R}\right) + \nu(R) \quad (21)$$